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CS-330

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Reflection & Design Decisions

For my final project, I recreated a scene from my pharmacy in a 3D environment using C++, OpenGL, and Visual Studio. I chose this idea because it's based on a real-world setting that I'm familiar with, and I felt like it would have enough interesting shapes that I could work with. My goal was to create a recognizable pharmacy workspace while using the modeling, texturing, lighting, and camera controls learned through this course.

The scene includes a countertop, also known as a pod, a counting tray, a spatula, pills, and a pill bottle. I used mostly basic shapes and combined them to create the objects.

I used shapes such as as boxes, cylinders, spheres, and prisms to build the objects. The spatula is made of several parts, including the blade, rounded end, gold details, handle, and pointed handle end. I also combined different shapes to create the pills and pill bottle.

For the counting tray, I needed a part that needed an opening on the top, like a hollow cylinder. I decided that I should create a custom mesh, as there wouldn't really be a good way to build the object otherwise. This ended up helping me with other shapes, like the pill bottle, which I used my hollow cylinder mesh for as well. I arranged everything on the pod to resemble my reference photo and make it as realistic as I could.

I used textures to make the scene look less plain. The countertop of the pod has its own texture, which I tiled over it, while the spatula has many different textures for its different components. The spatula blade uses a shiny metallic texture, a gold texture for the smaller details, and a wood texture for the handle.

Other objects use solid colors because they did not need extra textures, as they mostly are solid colors with a matte sheen to them. I wanted the textures to add detail without making the scene too busy.

I added a white overhead point light and a blue directional light. The white light was to provide general lighting for the scene, while to blue light adds some additional color and contrast.

My fragment shader uses ambient, diffuse, and specular lighting. These help control how bright objects appear, how light creates shading, and where highlights show up. I wanted the lighting to help make the objects look more three dimensional, especially the spatula.

I added the camera controls so users can explore the scene from different angles. The controls allow keyboard movement, WASD to move around with Q and E to move up and down, with mouse movement to look around, and scroll wheel adjustment of movement speed. Users can also switch between perspective, P, and orthographic, O, views.

These controls make it easier to look at objects and understand how they are arranged. The controls helped me a lot when creating the scene, so I could make sure everything lined up

correctly. Perspective gives the scene more depth, while orthographic projection provides a different view of the layout.

I really enjoyed this project. It helped me understand how shapes, textures, lighting and camera controls work together in a 3D scene. I learned how combining simple shapes can create more detailed objects and that textures and lighting can make them look even more realistic. This project even pushed me to figure out how to create my own mesh, which I am very proud of.

Overall, I wanted to recreate a scene from the pharmacy I work at that would be recognizable and demonstrated the techniques covered in this course. If I continued working on it, which I probably will, I would focus on adding more details and improving the appearance of the materials and lighting, and I would love to work on texturing and expanding everything in the scene.